

Australian Public Health Mortality Dashboard

A SQL-driven analysis of mortality trends across Australian states and territories, 2019–2023 — translating six SQL queries into a single view of where, why, and how prematurely Australians are dying.

SQL · SQLite

Source: AIHW MORT Books via data.gov.au

Scope: State/territory level, Persons

Static PDF companion — see [dashboard.html](#) for the interactive version

National deaths, 2023

183,131

Down 4.1% from the 2022 peak
(190,939)

Leading national cause of death

Coronary heart disease

10.1% of all deaths, 2019–2023

Widest health equity gap

15 years

NT median age at death: 67
vs SA: 83

NT premature death rate

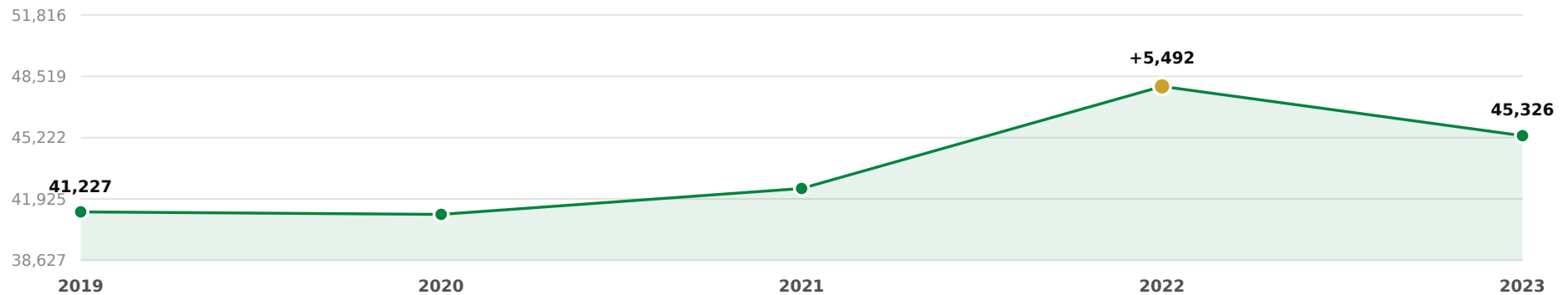
≈2× the ACT

329 vs 166 per 100,000,
population-adjusted

QUERY 1 · TREND

Victoria's mortality trend reveals a post-pandemic rebound, not a pandemic peak

Deaths in Victoria held steady through the 2020–2021 lockdown years, then jumped by 5,492 in 2022 — the sharpest year-on-year rise in the dataset.



Finding. Deaths remained relatively stable in 2020–2021 despite the pandemic, likely because strict lockdowns suppressed other causes of death (road accidents, influenza). The sharp 2022 spike (+13.4% vs 2021) reflects the COVID-19 rebound after restrictions lifted, compounded by healthcare deferred during the pandemic years.

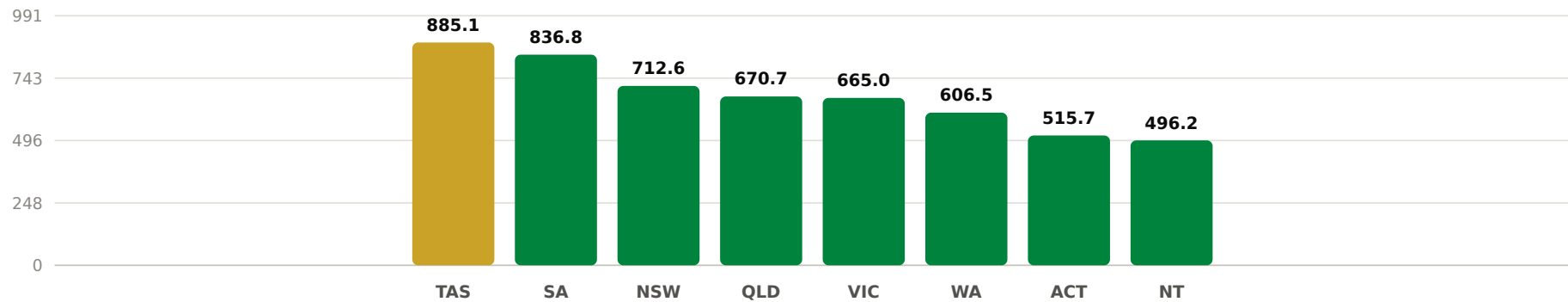
SQL QUERY

```
SELECT year, deaths
FROM mortality_summary
WHERE state = 'Victoria' AND sex = 'Persons'
ORDER BY year ASC;
```

QUERY 2 • RANKING

Crude death rate by state, 2023

Tasmania records the highest raw death rate in the country — but crude rates mix in age structure, so this ranking answers "how many died" per capita, not "how healthy is this state."



Finding. Tasmania leads at 885.1 deaths per 100,000 — driven by its older population structure — while the Northern Territory ranks last (496.2) despite having the country's worst health outcomes on other measures, because it has a much younger population. Crude rates alone are insufficient for comparing regions; they reflect age structure as much as health system performance.

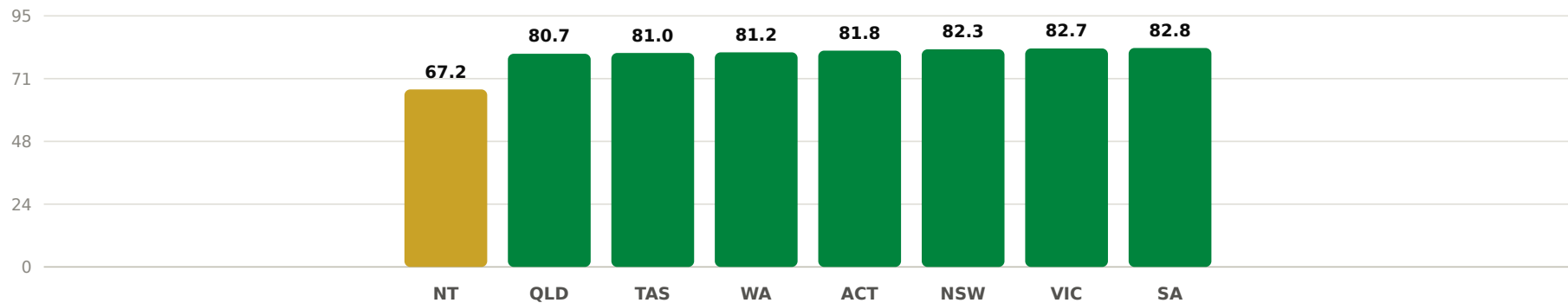
SQL QUERY

```
SELECT state, crude_rate_per_100000
FROM mortality_summary
WHERE year = 2023 AND sex = 'Persons'
ORDER BY crude_rate_per_100000 DESC;
```

QUERY 3 • HEALTH EQUITY

Average median age at death by state, 2019-2023

The starkest equity signal in the dataset: Territorians are, on average, dying 15 years younger than South Australians.



Finding. The Northern Territory has the lowest median age at death (67.2 years) — a full 15 years below South Australia (82.8). This is the clearest indicator of health inequity in the dataset, consistent with limited healthcare access and higher chronic disease burden in remote and Indigenous communities across the NT.

SQL QUERY

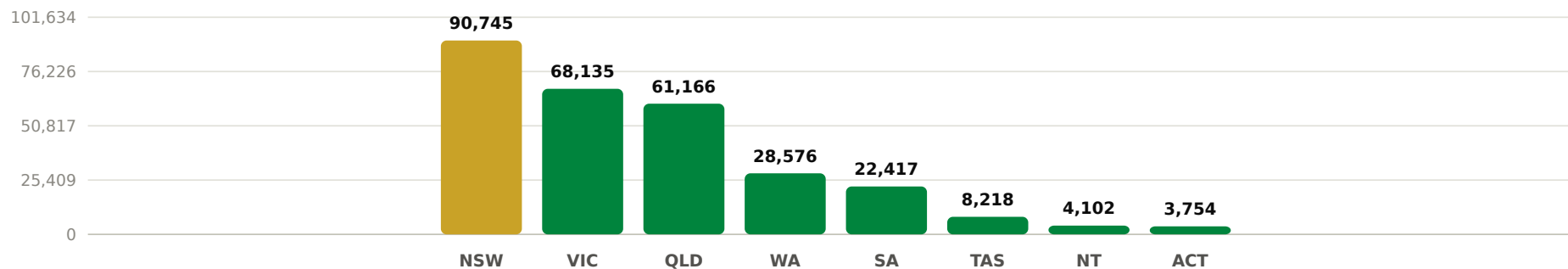
```
SELECT state, AVG(median_age) AS avg_median_age_at_death
FROM mortality_summary
WHERE sex = 'Persons'
GROUP BY state
ORDER BY avg_median_age_at_death ASC;
```

QUERY 4 & 5 · ABSOLUTE VS. POPULATION-ADJUSTED

Premature deaths, 2019-2023: raw counts mislead, adjusted rates don't

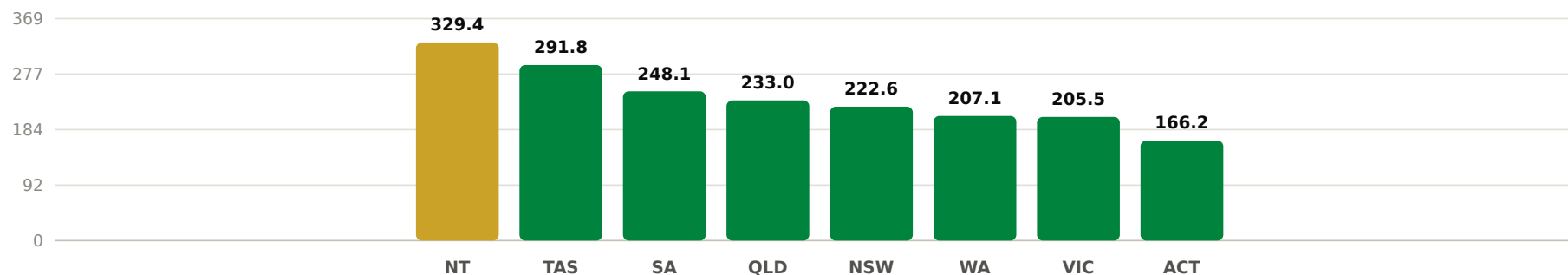
Premature death = death before age 75. Switch views to see why NSW looks like the biggest problem in absolute terms, but the Northern Territory is where the burden is actually heaviest.

ABSOLUTE TOTAL, 2019-2023



Finding (absolute). NSW (90,745) and Victoria (68,135) lead in absolute premature deaths — but this largely just reflects their much larger populations. Absolute counts alone don't identify where health intervention is most needed.

POPULATION-ADJUSTED RATE, AVG. 2019-2023



Finding (population-adjusted). Once controlled for population size, the Northern Territory (329 per 100,000) carries nearly **double** the premature mortality burden of the ACT (166) — and overtakes every other jurisdiction. This is the view that should drive resource-allocation decisions.

SQL QUERIES

```
-- Query 4: absolute total
SELECT state, SUM(premature_deaths) AS total_premature_deaths
FROM mortality_summary
WHERE sex = 'Persons'
GROUP BY state
ORDER BY total_premature_deaths DESC;

-- Query 5: population-adjusted rate
SELECT state,
       AVG((premature_deaths * 1.0 / population) * 100000) AS premature_death_rate_per_100k
FROM mortality_summary
WHERE sex = 'Persons'
GROUP BY state
ORDER BY premature_death_rate_per_100k DESC;
```

QUERY 6 · JOIN

Leading cause of death by state

Joining mortality summary with leading causes shows coronary heart disease dominates nationally — except where an older population profile pushes dementia to the top.

 **Coronary heart disease leads**  **Dementia / Alzheimer's leads (exception)**

NSW — New South Wales

Coronary heart disease

9.6% of all deaths, 2019–2023

#2 Dementia / Alzheimer's
disease (9.2%)

#3 Cerebrovascular disease
(5.9%)

VIC — Victoria

Coronary heart disease

9.7% of all deaths, 2019–2023

#2 Dementia / Alzheimer's
disease (8.4%)

#3 Cerebrovascular disease
(5.3%)

QLD — Queensland

Coronary heart disease

10.8% of all deaths, 2019–2023

#2 Dementia / Alzheimer's
disease (9.2%)

#3 Cerebrovascular disease
(5.7%)

WA — Western Australia

Coronary heart disease

10.9% of all deaths, 2019–2023

#2 Dementia / Alzheimer's
disease (9.4%)

#3 Cerebrovascular disease
(5.1%)

SA — South Australia

Dementia / Alzheimer's disease

11.0% of all deaths, 2019–2023

-
- #2 Coronary heart disease (10.2%)
 - #3 Cerebrovascular disease (5.0%)

TAS — Tasmania

Coronary heart disease

11.2% of all deaths, 2019–2023

-
- #2 Dementia / Alzheimer's disease (8.2%)
 - #3 COPD (5.6%)

ACT — Australian Capital Territory

Dementia / Alzheimer's disease

10.2% of all deaths, 2019–2023

-
- #2 Coronary heart disease (8.4%)
 - #3 Cerebrovascular disease (5.3%)

NT — Northern Territory

Coronary heart disease

9.8% of all deaths, 2019–2023

-
- #2 Diabetes (7.6%)
 - #3 Lung cancer (5.9%)

Finding. Coronary heart disease is the leading cause of death in 6 of 8 states and territories. South Australia and the ACT are the exceptions — both show Dementia/Alzheimer's disease ranking first, consistent with their older population profiles where people survive long enough to develop degenerative conditions. The Northern Territory's #2 and #3 causes (diabetes, lung cancer) diverge from the national pattern (dementia, cerebrovascular disease) entirely — another marker of a distinct health profile.

SQL QUERY

```
SELECT m.state,
       AVG((m.premature_deaths * 1.0 / m.population) * 100000) AS premature_death_rate_per_100k,
       l.cause_of_death AS leading_cause_of_death
FROM mortality_summary m
JOIN leading_causes l ON m.state = l.state
WHERE m.sex = 'Persons' AND l.sex = 'Persons' AND l.rank = 1
GROUP BY m.state, l.cause_of_death
ORDER BY premature_death_rate_per_100k DESC;
```

SYNTHESIS

Summary of insights

FINDING	IMPLICATION FOR PUBLIC HEALTH POLICY
NT median age at death is 67 — 15 years below SA	Severe health inequity; the strongest case in the dataset for targeted intervention in remote and Indigenous health services.
NT premature death rate is roughly double the ACT's	Population-adjusted metrics reveal a burden that absolute counts hide entirely — resourcing decisions based on raw numbers would miss the NT.
2022 mortality spike in Victoria (+13.4% vs 2021)	A post-pandemic rebound effect; deferred care during lockdowns had measurable, delayed consequences.
Coronary heart disease leads in 6 of 8 states/territories	Cardiovascular prevention programs have consistent national relevance and scale well across jurisdictions.
Tasmania has the highest crude rate but a mid-range median age	A high crude rate mainly reflects an older population, not necessarily worse health system performance — always pair crude rates with an age-adjusted view.

▼ Full underlying data table (mortality_summary, Persons, 2019–2023)

STATE	YEAR	DEATHS	POPULATION	CRUDE RATE /100K	AGE-STD. RATE /100K	PREMATURE DEATHS	AVOIDABLE DEATHS	MEDIAN AGE
NSW	2019	56,042	8,046,748	696.5	531.5	18,433	8,883	82.1
NSW	2020	52,485	8,110,610	647.1	485.7	17,281	8,388	82.1
NSW	2021	56,525	8,097,062	698.1	509.5	17,860	8,323	82.4
NSW	2022	62,980	8,166,704	771.2	553.4	19,076	8,795	82.7
NSW	2023	59,451	8,342,285	712.6	512.3	18,095	8,409	82.4
VIC	2019	41,227	6,537,305	630.6	500.8	13,092	6,524	82.6
VIC	2020	41,093	6,615,046	621.2	485.2	13,087	6,478	82.6

STATE	YEAR	DEATHS	POPULATION	CRUDE RATE /100K	AGE-STD. RATE /100K	PREMATURE DEATHS	AVOIDABLE DEATHS	MEDIAN AGE
VIC	2021	42,486	6,547,822	648.9	491.7	13,410	6,260	82.7
VIC	2022	47,978	6,630,631	723.6	541.7	14,591	6,828	82.9
VIC	2023	45,326	6,815,441	665.0	500.1	13,955	6,776	82.5
QLD	2019	32,470	5,088,847	638.1	535.6	11,982	6,145	80.4
QLD	2020	31,367	5,165,613	607.2	501.0	11,547	5,867	80.3
QLD	2021	33,858	5,215,814	649.1	521.5	12,139	6,038	80.7
QLD	2022	38,160	5,320,941	717.2	564.4	12,854	6,355	81.1
QLD	2023	36,622	5,460,420	670.7	524.0	12,644	6,331	80.8
WA	2019	15,042	2,659,625	565.6	491.2	5,465	2,885	81.1
WA	2020	14,993	2,712,912	552.7	468.2	5,598	2,934	80.7
WA	2021	15,891	2,749,365	578.0	476.0	5,562	2,888	81.4
WA	2022	17,299	2,791,794	619.6	500.1	5,969	3,087	81.3
WA	2023	17,475	2,881,227	606.5	486.1	5,982	2,976	81.5
SA	2019	13,913	1,767,395	787.2	527.8	4,390	2,151	82.9
SA	2020	13,607	1,790,355	760.0	504.7	4,297	2,061	82.6
SA	2021	14,494	1,802,601	804.1	521.1	4,498	2,139	83.1
SA	2022	15,455	1,821,215	848.6	541.2	4,626	2,067	82.9
SA	2023	15,499	1,852,284	836.8	532.5	4,606	2,013	82.7
TAS	2019	4,662	547,841	851.0	586.9	1,650	843	80.7
TAS	2020	4,435	557,578	795.4	539.8	1,580	828	80.8
TAS	2021	4,769	567,239	840.7	559.9	1,636	843	81.1
TAS	2022	5,142	571,051	900.4	585.8	1,771	859	80.7
TAS	2023	5,073	573,156	885.1	563.2	1,581	775	81.5

STATE	YEAR	DEATHS	POPULATION	CRUDE RATE /100K	AGE-STD. RATE /100K	PREMATURE DEATHS	AVOIDABLE DEATHS	MEDIAN AGE
ACT	2019	2,033	435,730	466.6	462.7	659	323	82.6
ACT	2020	2,162	444,903	485.9	472.9	739	381	81.6
ACT	2021	2,207	452,508	487.7	464.7	761	372	81.6
ACT	2022	2,563	456,915	560.9	518.8	854	413	81.8
ACT	2023	2,406	466,566	515.7	473.4	741	348	81.6
NT	2019	1,142	246,559	463.2	742.1	776	476	66.2
NT	2020	1,141	247,428	461.1	690.5	792	460	67.4
NT	2021	1,211	248,151	488.0	721.4	795	454	67.2
NT	2022	1,338	250,228	534.7	763.4	883	522	67.8
NT	2023	1,253	252,529	496.2	671.9	856	480	67.2

SQL concepts demonstrated

SELECT / WHERE / ORDER BY

AND (multi-condition filters)

Aggregate functions — SUM, AVG

GROUP BY

Calculated columns (AS alias)

JOIN across two tables

Data source: AIHW Mortality Over Regions and Time (MORT) Books, 2019–2023, via data.gov.au. Scope limited to state/territory level by design, per the project's original analytical focus. "Other Territories" and the national total row are excluded from state comparisons above for readability but remain in the underlying data.

SQL analysis, dataset, and full documentation: github.com/cesaralbertopazo-cyber/Public-Health-Mortality-SQL

Cesar Pazo — Data Analyst, Melbourne, Australia. Palette uses Australia's official national colours (Pantone 348C green, Pantone 116C gold), stepped for on-screen legibility per WCAG contrast and colour-vision-deficiency guidelines.

